
O-RAN Work Group 7 White-box Hardware Hardware Architecture, Requirements and Reference Design Specification for Enterprise Microcell

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Foreword

This Technical Specification (TS) has been produced by O-RAN Alliance.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The contents of the present document are subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.

yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.

zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

The present document specifies system requirements, high-level architecture and hardware reference design for the Enterprise Microcell deployment scenario as specified in the Deployment Scenarios and Base Station Classes document[1].

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

- [1] O-RAN.WG7.DSC.0-v04.00, "O-RAN Deployment Scenarios and Base Station Classes 4.0".
- [2] 3GPP TR 21.905, "Vocabulary for 3GPP Specifications".
- [3] 3GPP TS 38.104, "NR; Base Station (BS) radio transmission and reception".
- [4] O-RAN.WG4.CUS.0-v10.00, "O-RAN Control, User and Synchronization Plane Specification 10.0".
- [5] O-RAN.WG4.MP.0-v10.00, "O-RAN Management Plane Specification 10.0".
- [6] 3GPP TS 38.113, "NR: Base Station (BS) Electromagnetic Compatibility (EMC)".
- [7] O-RAN.WG1.O-RAN-Architecture-Description-v07.00, "O-RAN Architecture Description 7.0".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist

the user with regard to a particular subject area.

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms and definitions given in [2] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in [2].

Carrier Frequency: Center frequency of the cell.

F1 interface: The open interface between O-CU and O-DU defined by 3GPP TS 38.473.

Frequency Band: A designated frequency range for the operation of the base station and the UE radios. 5G NR frequency bands are divided into two different frequency ranges: Frequency Range 1 (FR1) that mainly includes sub-6GHz frequency bands, some of which are bands traditionally used by previous standards but has been extended to cover potential new spectrum offerings from 410MHz to 7125MHz; Frequency Range 2 (FR2) that includes frequency bands from 24.25 GHz to 52.6 GHz. Bands in this millimeter wave range have shorter range but higher available bandwidth than bands in the FR1.

Frequency Range: It refers to bandwidth defined by the frequency range within which the Base Station can be operated, defined by the band-pass filter of the BS; e.g., 3.4 – 3.8 GHz (400 MHz)

Occupied Bandwidth (OBW): It refers to the bandwidth occupied by the base station when operated, defined by the sum of the active bandwidths of the band allocation(s) operated. As defined by 3GPP TS 34.121 section 5.8, occupied bandwidth is the bandwidth containing 99% of the total integrated power of the transmitted spectrum, centered on the assigned channel frequency. The bandwidth between the 0.5% power frequency points is the occupied bandwidth.

Instantaneous Bandwidth (IBW): It refers to the bandwidth in which all frequency components can be simultaneously analyzed. It is defined by the frequency boundaries of the operating band(s).

Fronthaul Gateway (FHGW): A fronthaul gateway is a physical entity that is located between a distributed unit and one or more radio units where it distributes, aggregates, and/or converts fronthaul protocols between the distributed unit and multiple radio units.

gNB: A RAN node providing NR user plane and control plane protocol terminations towards the UE, and connected via the NG interface to the 5GC

Integrated architecture: In the integrated architecture, the O-RU and O-DU are implemented on one platform. Each O-RU and RF front end is associated with one O-DU. They are then aggregated to O-CU and connected by F1 interface.

Split architecture: The O-RU and O-DU are physically separated from one another in this architecture. A switch may aggregate multiple O-RUs to one O-DU. O-DU, switch and O-RUs are connected by the fronthaul interface as defined in WG4.

3.2 Symbols

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in [2] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, as in [2].

7-2x	Fronthaul interface split option as defined by O-RAN WG4, also referred to as 7-2
3GPP	Third Generation Partnership Project

5G	Fifth-Generation [Mobile Communications]
5GC	5G Core
ACS	Adjacent Channel Selectivity
ADC	Analog to Digital Converter
AFE	Analog Front End
AFU	Antenna Filter Unit
ASIC	Application Specific Integrated Circuit
BH	Backhaul
BMC	Baseboard Management Controller
BPSK	Binary Phase Shift Keying
BS	Base Station
CFR	Crest Factor Reduction
CISPR	International Special Committee on Radio Interference
CLK	Clock
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
CU	Central Unit
DAC	Digital to Analog Converter
DDC	Digital Down Conversion
DDR	Double Data Rate
DFE	Digital Front End
DIMM	Dual-Inline-Memory-Modules
DL	Downlink
DPD	Digital Pre-Distortion
DSP	Digital Signal Processor
DU	Distributed Unit
DUC	Digital Up Conversion
eCPRI	Enhanced Common Public Radio Interface
ECC	Error Correcting Code
EMC	ElectroMagnetic Compatibility
EVM	Error Vector Magnitude
FCC	Federal Communications Commission
FEC	Forward Error Correction
FFT	Fast Fourier Transform
FH	Fronthaul
FHGW	Fronthaul Gateway
FPGA	Field Programmable Gate Array

GbE	Gigabit Ethernet
GNSS	Global Navigation Satellite System
GPP	General Purpose Processor
GPS	Global Positioning System
IEEE	Institute of Electrical and Electronics Engineers
IFFT	Inverse Fast Fourier Transform
IMD	InterModulation Distortion
IP	Internet Protocol
I/O	Input/Output
JTAG	Joint Test Action Group
L1	Layer 1, also referred to as PHY or Physical Layer
L2	Layer 2, also referred to as Data Link layer in OSI model
L3	Layer 3, also referred to as Network layer in OSI model
LDPC	Low-Density Parity Codes
LDO	Low dropout
LLS	Lower Layer Split
LRDIMM	Load-Reduced Dual In-line Memory Module
LNA	Low Noise Amplifier
LTE	Long Term Evolution
LVDS	Low-Voltage Differential Signaling
MAC	Media Access Control
MIMO	Multiple Input Multiple Output
MCP	Multi-Chip Package
NG	Next Generation
NR	New Radio
OAM	Operations, Administrations and Maintenance
O-CU	O-RAN Centralized Unit as defined by O-RAN
O-DU	O-RAN Distributed Unit as defined by O-RAN
O-DU _x	A specific O-RAN Distributed Unit having fronthaul split option x where x may be 6, 7-2x (as defined by WG4) or 8
O-RU	O-RAN Radio Unit as defined by O-RAN
O-RU _x	A specific O-RAN Radio Unit having fronthaul split option x, where x is 6, 7-2x (as defined by WG4) or 8, and which is used in a configuration where the fronthaul interface is the same at the O-DU _x
OCXO	Oven Controlled Crystal Oscillator
ORx	Observation Receiver
PA	Power Amplifier
PCIe	Peripheral Component Interface Express

PDCP	Packet Data Convergence Protocol
PHY	Physical Layer, also referred as L1
PLL	Phase Locked Loop
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
QSFP	Quad Small Form-factor Pluggable
RAN	Radio Access Network
RF	Radio Frequency
RLC	Radio Link Controller
RRC	Radio Resource Controller
RU	Radio Unit as defined by 3GPP
RX	Receiver
SATA	Serial ATA
SDU	Service Data Unit
SFP	Small Form-factor Pluggable
SFP+	Small Form-factor Pluggable Transceiver
SoC	System on Chip
SPI	Serial Peripheral Interface
SSD	Solid State Drive
TCXO	Temperature Compensated Crystal Oscillator
TDP	Thermal Design Power
TR	Technical Report
TS	Technical Specification
TX	Transmitter
UL	Uplink
USB	Universal Serial Bus
WG	Working Group

4 Deployment Scenarios and White Box Base Station Architecture

This chapter consists of two parts: the deployment scenario and the white box architecture. The deployment scenarios outline more specific functional requirements of the base station. All the reference designs shall meet these requirements in order to comply with O-RAN white box standard. In the white box hardware architecture section, it describes the overall gNB hardware architecture and function partition that meet the design requirements. The details on each of these topics are described in the following sections.

4.1 Deployment Scenarios

The specification here addresses a specific deployment scenario defined by the white box Deployment Scenarios and Base Station Classes document [1]. Its requirements are also defined in [1]. Some of the key requirements described in that document are highlighted below:

- Cell type: Enterprise Microcell
- Mounting Options: Rooftop, side of building/wall, pole
- Possible Carrier Frequency Bands: n41, n78
- [Instantaneous] Bandwidth: Up to 100 MHz
- Antennas Configuration: Up to 8T8R
- Conducted Power: up to 20W per port
- Fronthaul Type: O-RAN FH (WG4) split option 7-2x
- Architecture: Split architecture

4.2 White Box Base Station Architecture

In general, the base station hardware architecture can be classified by using different criteria. The physical partition method is adopted by O-RAN; hence, the base station architecture is divided into two categories namely split and integrated. In split architecture, the fronthaul interface determines the gNB functions location. Here we refer to the two partitions as O-DU_x and O-RU_x; where the “x” is split option number. In case the O-RAN WG4 defined fronthaul interface is used, these two partitions are called O-DU_{7-2x} and O-RU_{7-2x}. For a more complete description of the terminology used, refer to the Deployment Scenarios and Base Station Classes Document [1].

For this implementation of Enterprise Microcell, the split architecture is used. With split architectures, the O-CU and O-DU_x are integrated into one Whitebox, as shown in Figure 4.2-1. Optionally, a second O-RU may be cascaded to the first O-RU_x via the FH interface. The FH interface between the two cascaded O-RUs shall be the same as the FH interface between the O-DU_x and the first O-RU_x.

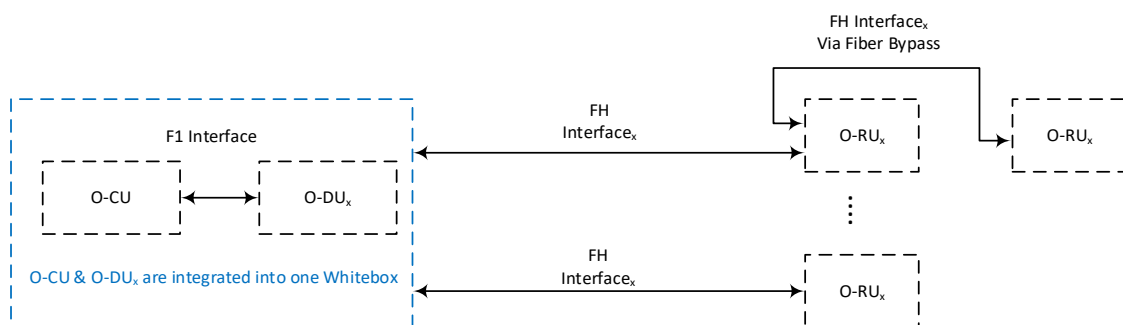


Figure 4.2-1 Split Architecture

5 Hardware Architecture

In the white box hardware architecture section, it describes the overall gNB hardware architecture and function partition that meet the design requirements. The details on each of these topics are described in the following sections.

5.1 O-CU Hardware Architecture

In a 3GPP system architecture, the gNB Central Unit (CU) communicates to the Distribution Unit (DU) via an F1 interface. This interface has been adopted by O-RAN Alliance for O-CU and O-DU_x connection. F1 is an IP based protocol interface, which offers more flexibility in the O-CU platform design.

5.1.1 O-CU Architecture Diagram

The O-CU can be implemented with any General-Purpose Processor (GPP) based platform having an optional hardware accelerator block. The O-CU functions can be implemented either on separated hardware platforms or on the same hardware platforms integrated with O-DU_x functions. In either case, the O-CU hosts L2/L3 functions, whereas O-DU hosts L2/L1 functions which require different CPU, Storage, Acceleration and Networking capabilities. Figure 5.1-1 shows the hardware blocks and interfaces within the O-CU white box.

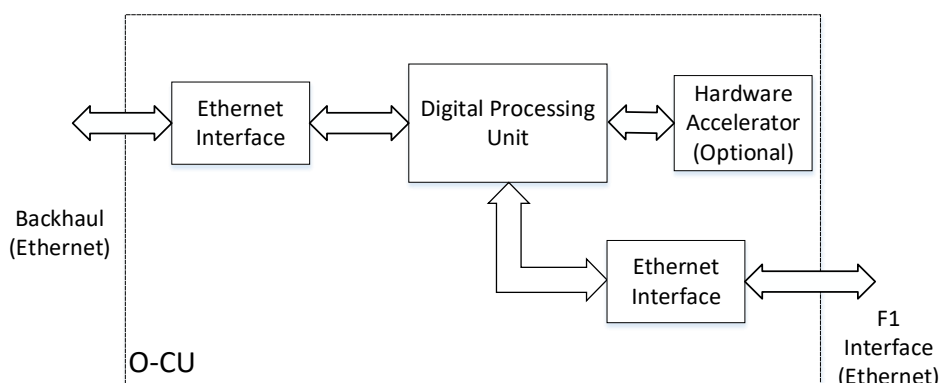


Figure 5.1-1 O-CU White Box Hardware Block Diagram

5.1.2 O-CU Functional Module Description

The O-CU functional architecture comprises Digital Signal Processing, optional HW Acceleration, and Connectivity units as well as memory and storage units as shown in Figure 5.1-1.

5.1.3 O-CU Interfaces

The O-CU interfaces through backhaul with the 5GC core network and via F1 interface with the O-DU_{7-2x}. The backhaul and the F1 interfaces are typically implemented with GbE transport/connections.

5.2 O-DU_{7-2x} Hardware Architecture

For a split RAN architecture, the functional blocks of the RAN physical layer are divided into two parts – high PHY and low PHY. The O-RAN lower layer split is defined in O-RAN WG4 fronthaul interface specification [4]. It also provides the details of the interface protocol as well as function partitions.

5.2.1 O-DU_{7-2x} Architecture Diagram

Depending on the 3GPP standards and category of the radio unit, the split function blocks within O-DU_{7-2x} and O-RU_{7-2x} may vary accordingly. The O-RAN fronthaul C/U/S-plane specification [4] provides comprehensive information on this topic. The hardware functional partition architecture is shown in Figure 5.2-1.

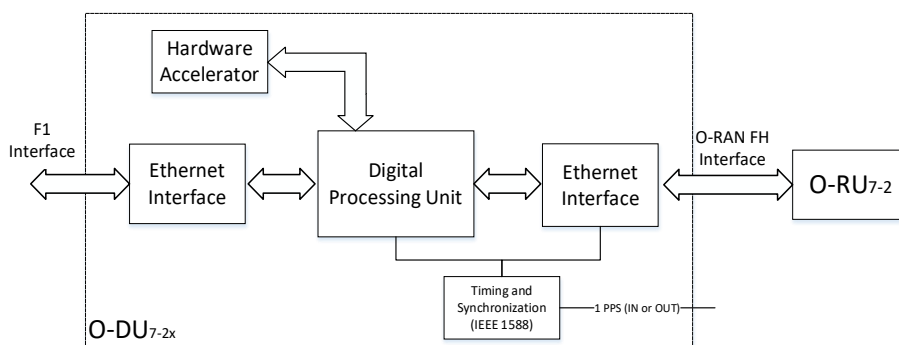


Figure 5.2-1 O-DU_{7-2x} with Split Physical Layer Function

5.2.2 O-DU_{7-2x} Functional Module Description

The choice of O-DU_{7-2x} hardware components depends on the product design requirements and is outside the scope of this specification. Figure 5.2-2 shows the main components of O-DU_{7-2x}. Their descriptions and requirements are as follows:

Digital Processing Unit

The processing unit can be any GPP or digital signal processor (DSP), with I/O chipset. It may also be in the form of System-On-Chip (SOC), or Multi-Chip Package (MCP).

Memory

DDR memory devices are used to store the runtime data and software for the processing unit.

Flash Memory

On board non-volatile storage device is used to store the firmware and non-volatile data, such as log data.

Board Management Controller

The controller is used to manage/control the power and monitors the operational status of the board.

Storage Device

The storage device such as hard drive is used to store OS, driver and applications software.

Ethernet Controller

The Ethernet ports transport the fronthaul or backhaul traffic according to the gNB hardware node requirements. The Ethernet device shall support IEEE1588 based timing synchronization.

Hardware Accelerator

The hardware accelerator is an optional device. For performance improvement, the hardware accelerator can be used to process computationally-intensive functions (e.g., Forward Error Correction (FEC)) and to offload the processor.

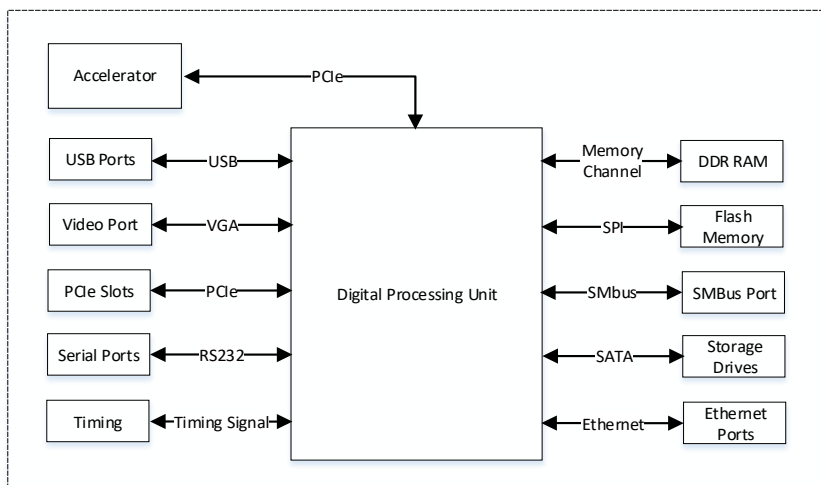


Figure 5.2-2 O-DU_{7-2x} Functional Block and Interface Diagram.

5.2.3 O-DU_{7-2x} Interfaces

The O-DU_{7-2x} supported interfaces described below are also shown in Figure 5.2-2.

Memory Channel Interface

Support DDR4 and later memory interface.

PCIe Interface

Support for PCIe Gen3 x16 and later interface versions; the bandwidth depends on the use cases, and it can be used to connect an accelerator device or network card.

Ethernet Interfaces

Supports any one or combination of GbE/10GbE/25GbE/40GbE links.

Serial ATA Interface

SATA3 shall be supported in case of software storage, such as hard drive.

SPI Interface

The SPI interface connects the processor with flash type of device for firmware.

Video Interface

Video interface is optional

USB Interface

Used to connect with local device for debug or on-board firmware update.

Miscellaneous Interface

Other interfaces that may be needed such as serial port, JTAG, etc.

5.3 O-RU_{7-2x} Hardware Architecture

This section provides the high-level architecture and functional module description of O-RU_{7-2x}.

5.3.1 O-RU_{7-2x} Architecture Diagram

O-RU_{7-2x} consists of five modules, namely fronthaul processing unit, digital processing unit, RF processing unit, timing unit, and power unit, as shown in Figure 5.3-1. The digital processing unit handles all digital signals, and the RF processing unit handling all analog signals. There is a data conversion block for each transceiver path after the digital processing block which converts between digital signals and analog signals. The FH processing unit complies with O-RAN WG4 open fronthaul interface. There is at least two interface port available, one for connecting with O-DU_{7-2x}, and the other for O-RU cascading.

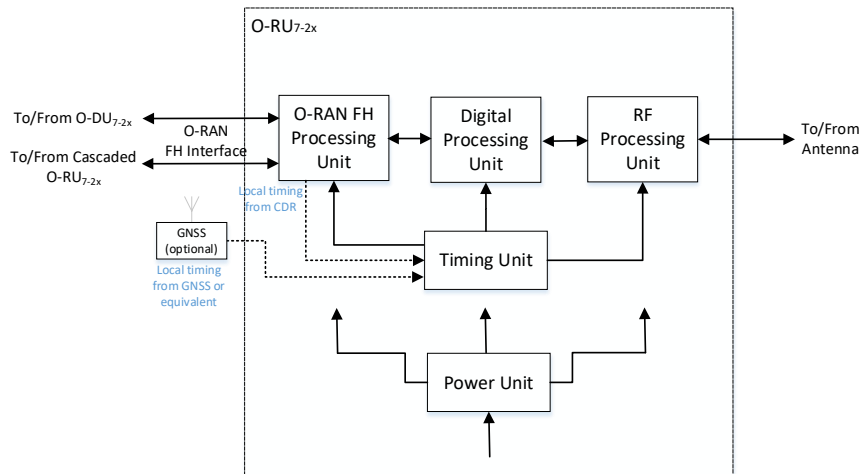


Figure 5.3-1 O-RU_{7-2x} Architecture Diagram

5.3.2 O-RU_{7-2x} Functional Module Description

Figure 5.3-2 illustrates O-RU_{7-2x} functional blocks that support O-RAN FH with split option 7-2x. The digital processing unit block of O-RU_{7-2x} is responsible for low-PHY functions such as FFT/iFFT, CP addition/removal, and PRACH filtering. Digital Down Converter (DDC), Digital Up Converter (DUC), Crest Factor Reduction (CFR) and Digital Pre-Distortion (DPD) functions are also performed in digital processing unit. For bandwidth reduction, O-RU_{7-2x} architecture also supports the optional compression and decompression functions of FH interface. The ADC and DAC are mixed signal devices responsible for conversion of data between the digital and analog domains. As such, this block can be included as part of the either the digital processing unit or the RF processing unit. The RF Processing Unit consists of an optional frequency converter (mixer), Power Amplifier (PA)/ Low Noise Amplifier (LNA) and TX/RX filters. An antenna module follows that comprises physical antennae, RF feed distribution/aggregation network, and calibration network. The timing unit includes any clock and frequency synthesis required as well as other timing and synchronization circuits. In any given implementation, the physical boundaries between blocks may be slightly different, but the logical interfaces will generally follow those shown in Figure 5.3-2.

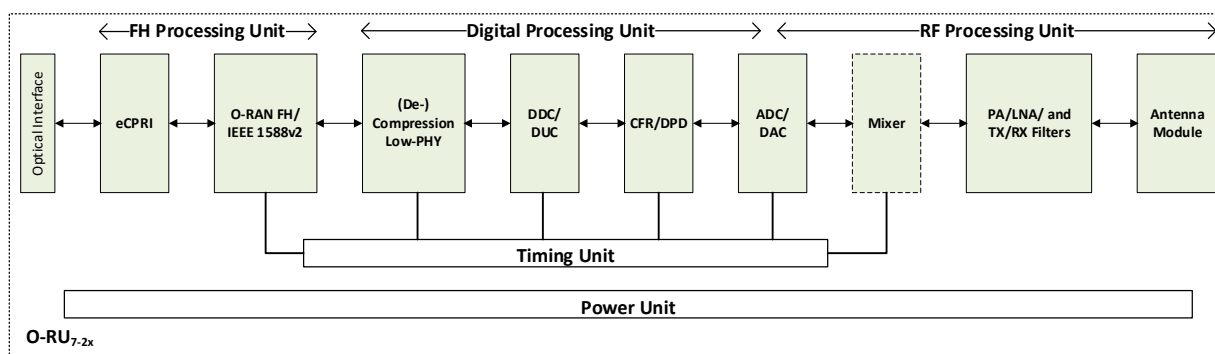


Figure 5.3-2 O-RU_{7-2x} Functional Module Diagram

6 Hardware Requirements

This chapter provides the requirements for various white boxes used within the Enterprise Microcell base station. These white boxes are O-CU, O-DU_x and O-RU_x. In this version of the specification, the O-CU and O-DU_x are implemented in an integrated fashion into one white box hardware. The reference design based on specification shall meet all these requirements based on their priority as specified by operators.

6.1 O-CU Requirements

The requirement for O-CU portion is captured in the following O-DU_x section.

6.2 O-DU_x Common Requirements

The common requirements apply to all the O-DU_x regardless of the split option. However, in this version of the specification, the focus will be on O-DU_{7-2x}. If more options get standardized in a future version of this document, some of these common requirements may be moved to the specific sub-sections in section 6.3.

6.2.1 Performance

The performance requirements of the O-DU_{7-2x} for Enterprise Microcell are listed in Table 6.2-1, including the channel bandwidth, antenna configuration, transmission distance, synchronization, etc.

Table 6.2-1 O-DU_{7-2x} Performance Requirements

Parameter	Requirement	Description	Priority
Channel Bandwidth	≤200 MHz OBW	RF Bandwidth	High
Antenna Configuration (Number of Transceivers)	up to 8T8R	Tx/Rx antenna number	High
Transmission distance	>2 km	Distance between O-DU and O-RU	High
Synchronization	Support 1588v2 Synchronization	Timing synchronization	High

	Support GPS Synchronization	method	High
	Support BeiDou Synchronization		High
	Support Galileo Synchronization		Low
	Support GLONASS Synchronization		Low
	Support switching between 1588v2 and GNSS		High
	Support SyncE		Low
	Sync. between BS $\leq \pm 1.5$ us; Carrier freq. error within one subframe $\leq \pm 0.05$ PPM		High
Capacity/MIMO	Support at least 4 100MHz bandwidth 4T4R cells	MIMO related capability	High
Peak Rate	With 100MHz bandwidth and 64% for DL, the peak throughput of one cell: DL: 1.5Gbps (4 layers, 256QAM) UL: 280Mbps (2 layers, 64QAM)	Peak data rate	High
Modulation	DL: QPSK, 16QAM, 64QAM, 256QAM UL: $\pi/2$ -bpsk, QPSK, 16QAM, 64QAM	Modulation schemes	High
	UL: 256QAM		Medium
Latency	Control Plane < 10 ms (def: Msg1 to Msg5) User Plane DL < 4 ms UL < 4 ms (def: PDCP SDU- PDCP SDU)	Control/user plane Latency	High

6.2.2 Interfaces

The interface requirements of the O-DU_{7-2x} are listed in Table 6.2-2.

Table 6.2-2 O-DU_{7-2x} Interface Requirements

Parameter	Requirement	Description	Priority
Transport Interface	10 GbE or 25GbE NG interface to connect with 5G core	transport links	High
	At least 4 fronthaul interfaces connected with O-RU _{7-2x}		High

6.2.3 Environmental and EMC

The EMC requirements of the O-DU_{7-2x} are listed in Table 6.2-3.

Table 6.2-3 O-DU_{7-2x} EMC Requirements

Parameter	Requirement	Description	Priority
EMC	Complying with the requirements of 3GPP TS 38.113 (2017-12R15) [6] for equipment used in telecommunication room	Electromagnetic Compatibility requirement	High

6.2.4 Mechanical, Thermal and Power

The mechanical requirements of the O-DU_{7-2x} are listed in Table 6.2-4.

Table 6.2-4 O-DU_{7-2x} Mechanical Requirements

Parameter	Requirement	Description	Priority
Dimension	Built in any 19" standard rack, or stand alone, and the height does not exceed 2U, the depth (including the connector) must be less than 750mm.	Measurement in three dimensions	High
Status Indicator LED	At least include the following status indicators: ---1 indicating on/off status of the power supply ---1 indicating on/off status of the transmission link	Indicator lights	High

The thermal requirements of the O-DU_{7-2x} are listed in Table 6.2-5.

Table 6.2-5 O-DU_{7-2x} Thermal Requirements

Parameter	Requirement	Description	Priority
Temperature and Humidity	The O-DU _{7-2x} shall operate steadily and reliably over a long period of time under the following environmental conditions for data center: Operating Temperature: -5 °C ~ + 55 °C Relative humidity: 15% ~ 85%	Environmental requirements for reliability	High
	The solution should support extended temperature range (-40°C to +65 °C) and humidity (5%-95%) if implemented outdoors.	Environmental requirements for reliability	Low

The power requirements of the O-DU_{7-2x} are listed in Table 6.2-6.

Table 6.2-6 O-DU_{7-2x} Power Requirements

Parameter	Requirement	Description	Priority
Power Supply	DC -48 VDC (-40V ~ -57V) (can be connected to AC/DC converter) or AC 220V power supply, voltage range of 140V~300V, frequency range of 45Hz ~ 65Hz.	Power supply	High
Power Dissipation	< 480 W, with full load operation	Power Requirement	Medium

6.3 O-DU_x Split Option Specific Requirements

‘This section intentionally left blank’.

6.4 O-RU_x Requirements

The common requirements apply to all the O-RU_x regardless of the split option. However, in this version of the specification, the focus will be on O-RU_{7-2x}. If more options get standardized in a future version of this document, some of these common requirements may be moved to the specific sub-sections in section 6.5.

6.4.1 Performance

The O-RU_{7-2x} performance requirements cover all the aspects of radio unit including frequency bands, antenna configurations, power efficiency, etc. Table 6.4-1 lists the performance parameters related to O-RU_{7-2x} for Enterprise Microcell.

Table 6.4-1 O-RU_{7-2x} Performance Requirements

Parameter	Requirement	Description	Priority
Operating band	n78, n41	Radio frequency band	High
[Occupied] Bandwidth	≤200 MHz	Frequency bandwidth	High
Antenna Configuration (Number of Transceivers)	Up to 8T8R	Tx/Rx Antenna numbers	High
Output Power Accuracy	Under normal condition: ±2dB	Power accuracy	High
Tx off Power Level	less than -85dBm/MHz; See section 6.4.1 in 3GPP TS 38.104 [3].	Tx power level during an off period	High
EVM at maximum output power	64QAM: EVM smaller than 5% 256QAM: EVM smaller than 3.5%	Max output power	High
Operating band unwanted emissions	The Operating band unwanted emissions shall satisfy the Category B limit defined by the section 6.6.4.2.4 in 3GPP TS 38.104 [3].	RF operation band unwanted emissions requirement	High
Transmitter spurious emissions	The Operating band unwanted emissions shall satisfy the Category B limit defined by the section 6.6.4.2.4 in [3].	Deliberately generated RF signal by transmitter	High
Receiver Sensitivity	The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel of G-FR1-A1-5; the reference sensitivity levels shall be better than -95.6dBm.	The weakest signal the receiver can identify and process	High
Blocking	In Channel selection, ACS, In-band blocking, out-band blocking, IMD and other receiver specification shall follow the 3GPP guidelines in [3], the reference sensitivity is allowed to degrade at most 6dB under all kinds of interference signal and corresponding level.	Channel selection related requirement	High
Other specifications	Except for all the RF specifications listed above, other RF specifications shall follow the requirement in [3].	Additional standard to comply	High
Downlink modulation Mode	QPSK、16-QAM、64-QAM、256-QAM	DL Modulation schemes	High
Uplink modulation	$\pi/2$ -BPSK、QPSK、16-QAM、64-QAM	UL Modulation	High

mode	256QAM	schemes	Medium
Conducted Power	Up to 20W per port	RF power	High
EIRP	NA		

6.4.2 Interfaces

The interface related requirements of the O-RU_{7-2x} are listed in Table 6.4-2.

Table 6.4-2 O-RU_{7-2x} Interface Requirements

Parameter	Requirement	Description	Priority
Interface Number	The O-RU _{7-2x} shall have one fronthaul interface based on the split option supported	Number of fronthaul links	High
	The O-RU _{7-2x} shall have one optical interface for cascade	Number of transport links	High

6.4.3 Environmental and EMC

The environmental and EMC requirements of the O-RU_{7-2x} are listed in Table 6.4-3.

Table 6.4-3 O-RU_{7-2x} Environmental and EMC Requirements

Parameter	Requirement	Description	Priority
Mounting method	Rooftop, side of building/wall or pole	Mounting requirement	High
Grounding	The O-RU _{7-2x} shall support Joint Grounding Method and shall work normally when the grounding resistor is less than 10Ω.	Grounding requirement	High
EMC	The O-RU _{7-2x} shall comply with the requirements of 3GPP TS 38.113 [6].	Electromagnetic Compatibility requirement	High

6.4.4 Mechanical, Thermal and Power

The mechanical, thermal and power requirements of the O-RU_{7-2x} are listed in Table 6.4-4.

Table 6.4-4 O-RU_{7-2x} Mechanical, Thermal and Power Requirements

Parameter	Requirement	Description	Priority
Weight	The gross weight of the O-RU _{7-2x} shall be <12Kg	Weight requirement	High
Dimension/Volume	The volume of the O-RU _{7-2x} shall be <12L	Dimensional Measurements	High
Stability	The MTBF shall be >200000 hours	Mean Time Between Failures	High
Power Consumption	At full load, the power consumption shall not exceed 200W	Power requirement	High
Power supply	DC -48 VDC (-40v ~ -57v) (can be connected to AC/DC converter) or AC 220V power supply, voltage range of 140V~ 300V, frequency range of 45Hz ~ 65Hz.	Power Support Requirement	High
Level of protection	The protection level of the O-RU _{7-2x} shall be equivalent to the IP65 standard.	Protection level	High
Temperature and Humidity	The O-RU _{7-2x} shall operate at the temperature range of -40°C to +55°C with both cold start and hot start options. Humidity : 5%~95%	Ambient temperature and moisture requirement	High
Atmospheric pressure	The O-RU _{7-2x} shall operate normally under atmospheric pressure between 70 to 106Kpa.	Operational atmospheric pressure requirement	High
Cooling mode	Passive cooling	System cooling requirement	High

6.5 O-RUx Split Option Specific Requirements

‘This section intentionally left blank’.

7 Hardware Reference Design

This chapter describes one example of white box hardware reference design including O-CU, O-DU and O-RU.

7.1 O-CU Hardware Reference Design

The O-CU white box hardware is the platform that perform the O-CU function of upper L2 and L3. The hardware systems specified in this document meet the computing, power and environmental requirements of use cases configurations and feature sets of RAN physical node. These requirements are described in chapter 6 of this document. For enterprise scenarios, the UPF and/or MEC platform can also be deployed in the O-CU hardware in

order to meet various industrial requirements.

The O-CU hardware includes the chassis platform, mother board, peripheral devices, and cooling devices. The mother board contains processing unit, memory, the internal I/O interfaces, and external connection ports. The midhaul (MH) and backhaul (BH) interface are used to carry the traffic between O-CU and O-DU as well as O-CU and core network. The other hardware functional components include: the storage for software, hardware and system debugging interfaces, board management controller, just to name a few; the O-CU designer will make decision based on the specific needs of the implementation.

The HW reference design of O-CU is the same as O-DU except for the need of HW accelerator, thus detail design will be described in the next section.

7.2 O-DU_{7-2x} Hardware Reference Design 1

The O-DU_{7-2x} white box hardware is the platform that performs the O-DU_{7-2x} functions such as upper L1 and lower L2 functions. The hardware systems specified in this document meet the computing, power and environmental requirements of use case's configurations and feature sets of RAN physical node. These requirements are described in chapter 6 of this document. The O-DU_{7-2x} hardware includes the chassis platform, mother board, peripheral devices, and cooling devices. The mother board contains processing unit, memory, the internal I/O interfaces, and external connection ports. The fronthaul and backhaul interface are used to carry the traffic between O-RU_{7-2x} and O-DU_{7-2x} as well as O-CU and O-DU_{7-2x}. The O-DU_{7-2x} design may also provide an interface for hardware accelerator if that option is preferred. The other hardware functional components include: the storage for software, hardware and system debugging interfaces, board management controller, just to name a few. The O-DU_{7-2x} designer will make decisions based on the specific needs of the implementation.

Note that the O-DU_{7-2x} HW reference design is also feasible for O-CU and integrated O-CU/O-DU_{7-2x}.

7.2.1 O-DU_{7-2x} High-Level Functional Block Diagram

For this example, a dual-socket platform is used. In the following O-DU_{7-2x} mother board functional block diagram, it shows the interconnections of the major components and external interfaces. The digital processing unit or CPU handles the baseband processing workload. To make the processing more efficient, an accelerator can be used to assist with the baseband workload processing. The memory devices include the random-access memory (RAM) for temporary storage of data while flash memory is used for codes and logs. The storage device is for persistent storage. The external network cards can be used for fronthaul or backhaul connection. The baseband management controller (BMC) is a microcontroller which monitors hardware operation on motherboard.

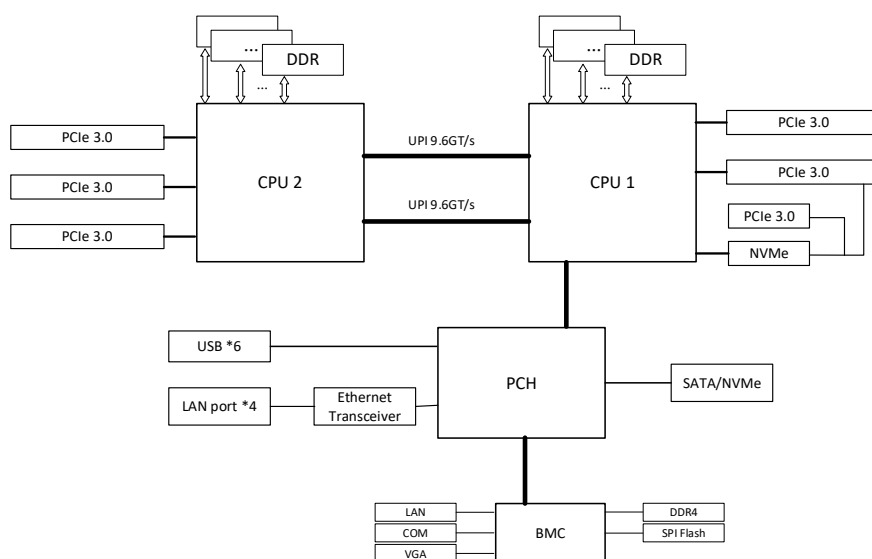


Figure 7.2-1 O-DU_{7-2x} Functional Block Diagram

7.2.2 O-DU_{7-2x} Hardware Components

7.2.2.1 Digital Processing Unit

The example of general-purpose processor performances and other related information are list in the following table.

Table 7.2-1 Processor Feature List

Item Name	Description
# of Cores	16
# of Threads	32
Base Frequency	2.30 GHz
Max Turbo Frequency	3.70 GHz
Cache	22 MB
TDP	110W
Max Memory Size (dependent on memory type)	1 TB
Memory Types	DDR4
Max # of Memory Channels	6
Sockets Supported	LGA-3647

The interface specifications on the main board are as follows:

Memory Channel Interfaces: The system memory capacity, type and related information is described in the following table.

Table 7.2-2 Memory Channel Feature List

Item Name	Description
Memory Types	DDR4 RDIMM/LRDIMM
# of Memory Channels	6
Support ECC	Yes
Maximum Memory Speed	2667 MHz
Memory Voltage	1.2 V

PCIe: PCIe Gen 3 is supported by the processor. There are total of 48 PCIe lanes for each processor.

Ethernet: The system provides 4*1Gbps Ethernet ports. When higher Ethernet bandwidth required, an Ethernet card can be installed in one of the PCIe slot.

7.2.2.2 Hardware Accelerator

Hardware accelerators can be used in O-DU_{7-2x} to offload computationally intensive functions and to optimize the performance under varying traffic and loading conditions. While the hardware acceleration functional

requirements and implementation are system designer's choice, the O-DU_{7-2x} is required to meet the minimum system performance requirements under various loading conditions and deployment scenarios.

For this example, the accelerator comprises a single FPGA for L1 offload and the fronthaul connectivity. The accelerator hardware and firmware features are listed in Table 7.2-3 and Table 7.2-4.

Table 7.2-3 Accelerator Hardware Features

SoC Resources	System Logic cells – 2,586 K CLB LUT – 1,182K CLB Flip-Flops -2,364K DSP Slices - 6,840 BRAM - 75.9Mb URAM - 270Mb
Form Factor	HHHL PCIe Form Factor
PCIe Interface	Gen 3 x16
Memory	8 GB DDR4
Fronthaul interface	4*10G/25G eCPRI interfaces
Timing Interface	SMA port for GNSS antenna; optical port for IEEE1588v2 synchronization; 2*RJ45 port for 1PPS&TOD (in & out)
Other External Interface(s)	Micro USB for JTAG support (FPGA programming and debug) and access to BMC
Board Management Controller	Telemetry, Security, Remote Upgrade
Operating Temperature (ambient)	-5°C to +55°C
Power	< 110 W

Table 7.2-4 Hardware Accelerator Firmware Features

Remote System Upgrade	Securely upgradable FPGA flash image
L1 Acceleration	5G NR LDPC encoding/decoding with interleaving/de-interleaving and rate-matching/rate-de-matching along with early termination, CRC attachment, and HARQ management
Fronthaul Protocols	Support O-RAN FH (WG4) split option 7-2x
Synchronization	Support synchronization and timing
Open Programmable Acceleration Environment	Support for: - FPGA flashing upgrade - Firmware version reporting - PCIe diagnostics - Ethernet diagnostics - Temperature and voltage telemetry information

Figure 7.2-2 illustrates the single-chip acceleration architecture comprising one FPGA with Gen3 x16 PCIe interfaces toward the Digital Processing Unit.

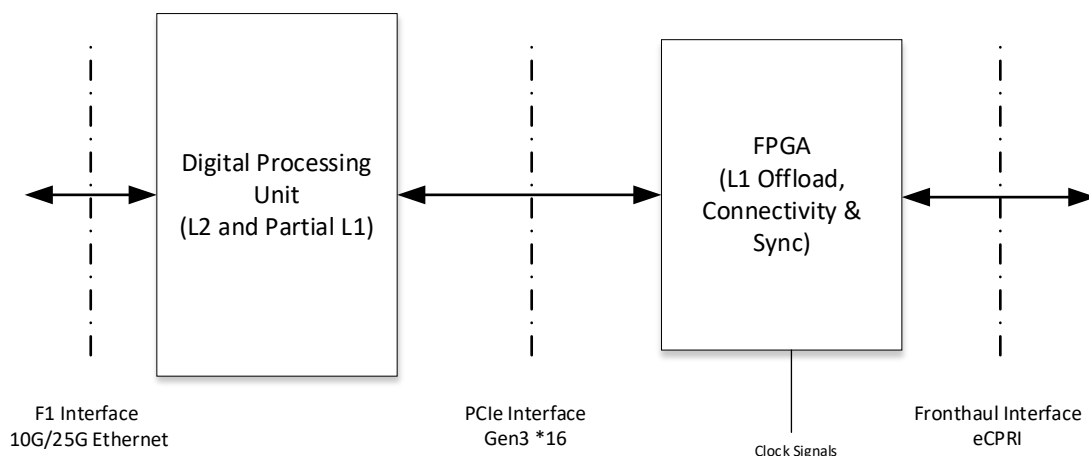


Figure 7.2-2 Single-chip FPGA-based Hardware Acceleration in O-DU_{7-2x}

7.2.3 Synchronization and Timing

This section describes the synchronization and timing mechanism that is used in the hardware accelerator.

Each hardware accelerator that supports connectivity functions (e.g., O-RAN fronthaul) in O-DU_{7-2x} must be able to support external synchronization I/O and to receive or transmit 1 PPS reference clock source in order to ensure synchronization across network.

The accelerator has a built-in DPLL, it can support GNSS, IEEE1588v2 or external reference signals as clock inputs, and output accurate frequency and synchronization signals to internal and external devices. When all reference inputs failed, the OCXO module can assist the accelerator to maintain high precision output signals for a certain period of time. Figure 7.2-3 describes the synchronization mechanisms of the accelerator.

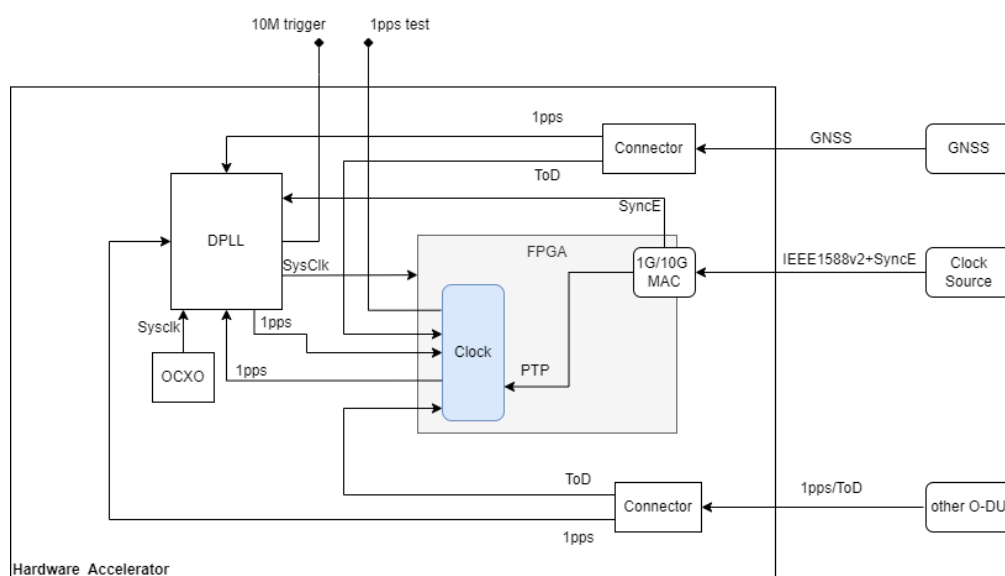


Figure 7.2-3 HW Accelerator Timing and Synchronization

7.2.4 External Interface Ports

The following table shows the external ports or slots that the system provided.

Table 7.2-5 External Port List

Port Name	Feature Description
Ethernet	1 x 1GbE RJ45 for management
	4 x 1GbE RJ45
PCIe	4 PCIe 3.0 x16 slots
	1 PCIe 3.0 x8 slots
	2 PCIe 3.0 x16 slots (with riser)
Storage	6 SAS/SATA/NVMe ports
USB	4 USB 3.0 ports
	2 USB 2.0 ports
VGA	1 VGA port
Serial Port	1 COM port

7.2.5 O-DU_{7-2x} Firmware

BIOS and BMC firmware are needed in the system and shall be installed.

7.2.6 Mechanical

7.2.6.1 Mother Board

The mechanical layout of the mother board shows the location of major components and interface ports.

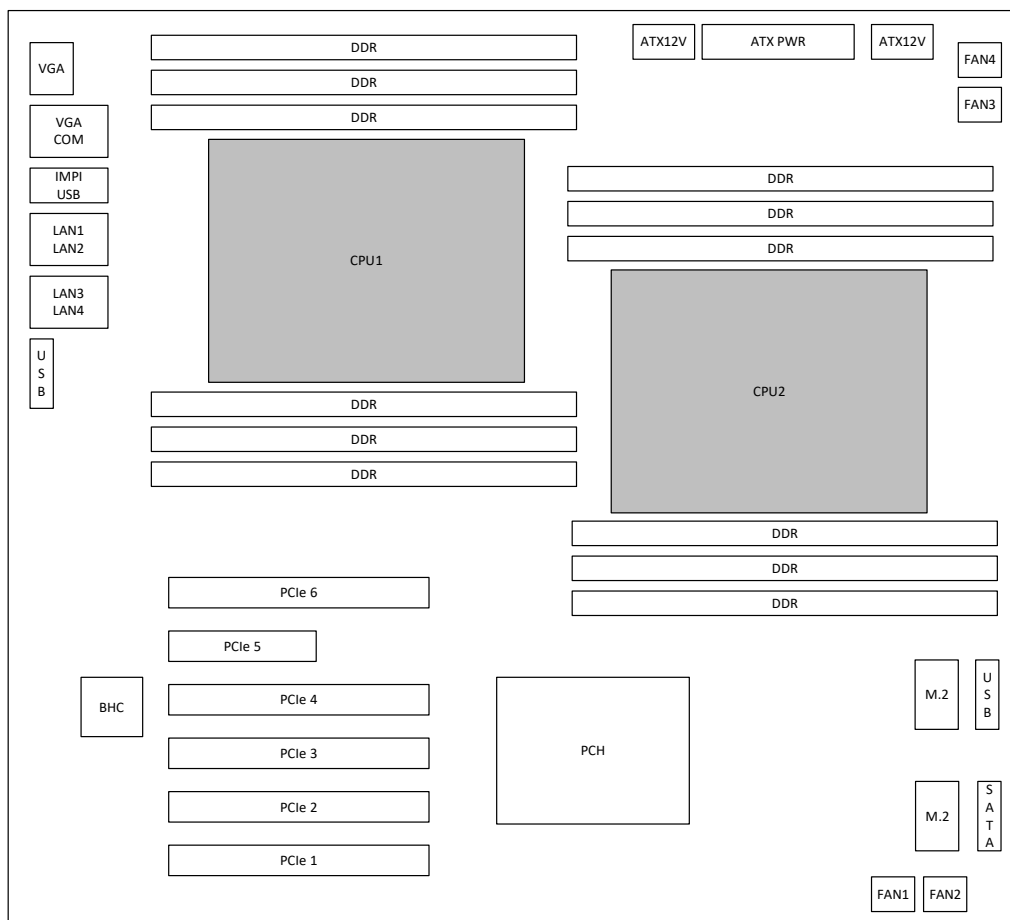


Figure 7.2-4 Mother Board Layout Diagram

7.2.6.2 Chassis

The 2U rack mount chassis contains the layout of the power supply, storage, and fans. The dimension of the hardware system is 450*446*87 (mm), and the weight is less than 25kg.

7.2.6.3 Cooling

The system installs 4x 80*38 PWM fans for the cooling.

7.2.7 Power Unit

In a fully loaded system, including the hardware acceleration card, the system power consumption should be less than 480W. The total system power shall be kept at less than 80% of the power supply capacity.

The power is provided by 1+1 CPRS 550W/800W (80Plus Platinum) power supply. Table 7.2-6 lists the power requirements of O-DU_{7-2x}.

Table 7.2-6 O-DU_{7-2x} Power Requirements

Supply Domain	Requirements
AC Input	100-240V, 50-60Hz
DC Input	240VDC
	Support -48V

7.2.8 Thermal

Active cooling with up to 4 fans is integrated in the chassis.

The hardware acceleration cards use passive cooling and a custom heatsink and is equipped with temperature sensors. It is designed to operate in temperatures ranging from -5°C to +55°C.

7.2.9 Environmental and Regulations

The O-DU_{7-2x} hardware system is RoHS Compliant. The power supply unit is EMC FCC/CISPR Class B compliant. Table 7.2-7 lists the environmental related features and parameters.

Table 7.2-7 Environmental Features

Item Name	Description
Operating Temperature	-5°C to +55°C
Non-operating Temperature	-40°C to 70°C
Operating Relative Humidity	8% to 90% (non-condensing)
Non-operating Relative Humidity	5% to 95% (non-condensing)

The hardware accelerator described in Section 7.2.2.2 is designed to operate in indoor environments and in temperatures ranging from -5°C to +55°C.

7.3 O-DU_{7-2x} Hardware Reference Design 2

7.3.1 O-DU_{7-2x} High-Level Functional Block Diagram

Figure 7.3-1 describes the components and connections inside the O-DU_{7-2x} white box. As described in the previous section, the O-DU_{7-2x} hardware can be implemented with difference design choices. Here, a system design is presented as an example which processes most of the O-DU_{7-2x} workload. The accelerator can be used to perform some O-DU_{7-2x} workload functions based on the overall performance requirement. Several Ethernet controllers are used for front haul link, back haul link and remote console control connection. The other parts include: RAM, flash memory, and hard drive storage. The USB ports are provided for hardware debug and local connection if needed. Finally, BMC block is mainly responsible for monitoring the hardware status.

The digital processing unit is a system which includes a 64-bit multi-core server class processor, integrated high-speed I/O, Integrated Memory Controllers (IMC), a PCIe Switch, and a BMC. The system supports double channel 128-bit wide processing instruction set. It also supports hardware virtualization to enable dynamic provisioning of services as communication service providers extend network functions virtualization (NFV). Figure 7.3-1 shows the major functional blocks of the digital processing unit.

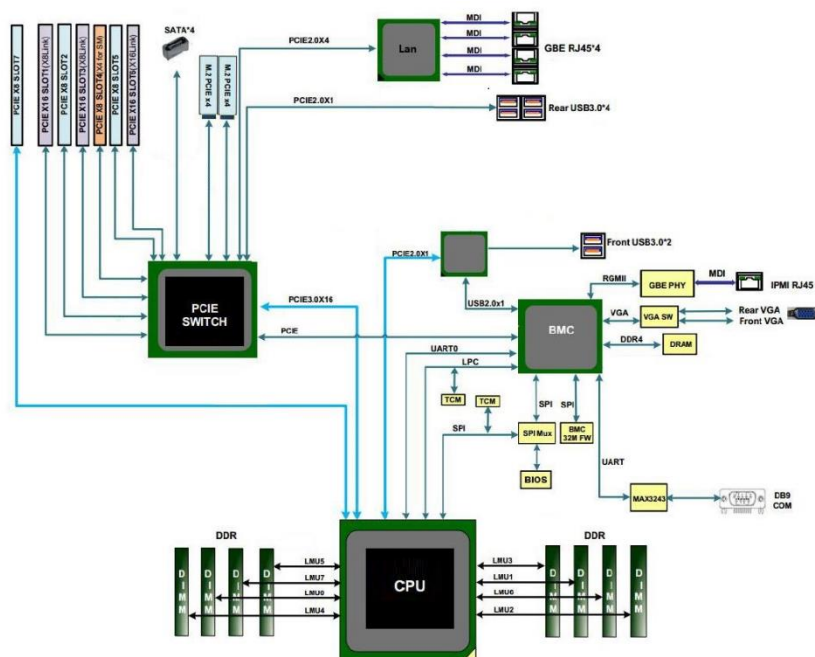


Figure 7.3-1 O-DU_{7-2x} Hardware Block Diagram

7.3.2 O-DU_{7-2x} Hardware Components

7.3.2.1 Digital Processing Unit

This example of the digital processing unit in O-DU_{7-2x} is based on the General Purpose Processor (GPP). The GPP requirements are listed in the following table.

Table 7.3-1 The Processor Feature List

Item Name	Description
# of Cores	64
Base Frequency	2.20 GHz
L2 Cache	32 MB
Thermal Design Power (TDP)	100W
Max Memory Size (dependent on memory type)	1TB
Memory Types	DDR4
Max # of Memory Channels	8

The interface specifications on the main board are as follows:

Memory Channel Interfaces: The system memory capacity, type and related information are described in the following table.

Table 7.3-2 The Memory Channel Feature List

Item Name	Description
Memory Types	DDR4
# of Memory Channels	8
Max Memory Speed	3200MHz

PCIe: PCIe Gen 3 should be supported by the processor. There are total of 33 PCIe lanes.

7.3.2.2 Hardware Accelerator

Refer to section 7.2.2.2.

7.3.2.3 Baseboard Management Controller

BMC is used to perform hardware power control (power on, power off and power cycle), monitor hardware status (temperatures, voltages, etc.), monitor Basic I/O System (BIOS)/ Unified Extensible Firmware Interface (UEFI) firmware status, and log system events. It provides remote access via shared or dedicated NIC. System user can do console access via serial or physical/Kernel-based Virtual Machine (KVM). The BMC has dedicated RAM and flash memory. It provides access via serial port or Ethernet port. Figure 7.3-1 describes the BMC connections with related components.

7.3.3 Synchronization and Timing

Refer to section 7.2.3.

7.3.4 External Interface Ports

The following table shows the external ports or slots that the system provided.

Table 7.3-3 External Port List

Item Name	Description
Ethernet	4x Gigabit Ethernet LAN connectors
	1GbE IPMI RJ45 port
	Dual 10GbE SFP or Dual 25GE SFP28 Faber Ethernet connectors
	6x25G SFP28 or 6x10G SFP
USB	5 USB 3.0 ports
Serial Port	1 DB9 COM port
Antenna port	1 SMA connector for GNSS
VGA	1 VGA port

7.3.5 O-DU_{7-2x} Firmware

BIOS and BMC firmware are needed in the system and shall be installed.

7.3.6 Mechanical

The mechanical design for mother board, chassis, and cooling are listed in this section.

7.3.6.1 Mother Board

The mechanical layout of the mother board shows the location of major components and interface ports. The following diagram also provides the dimension of the board is standard E-ATX/EEB/CEB (13*12 inch).

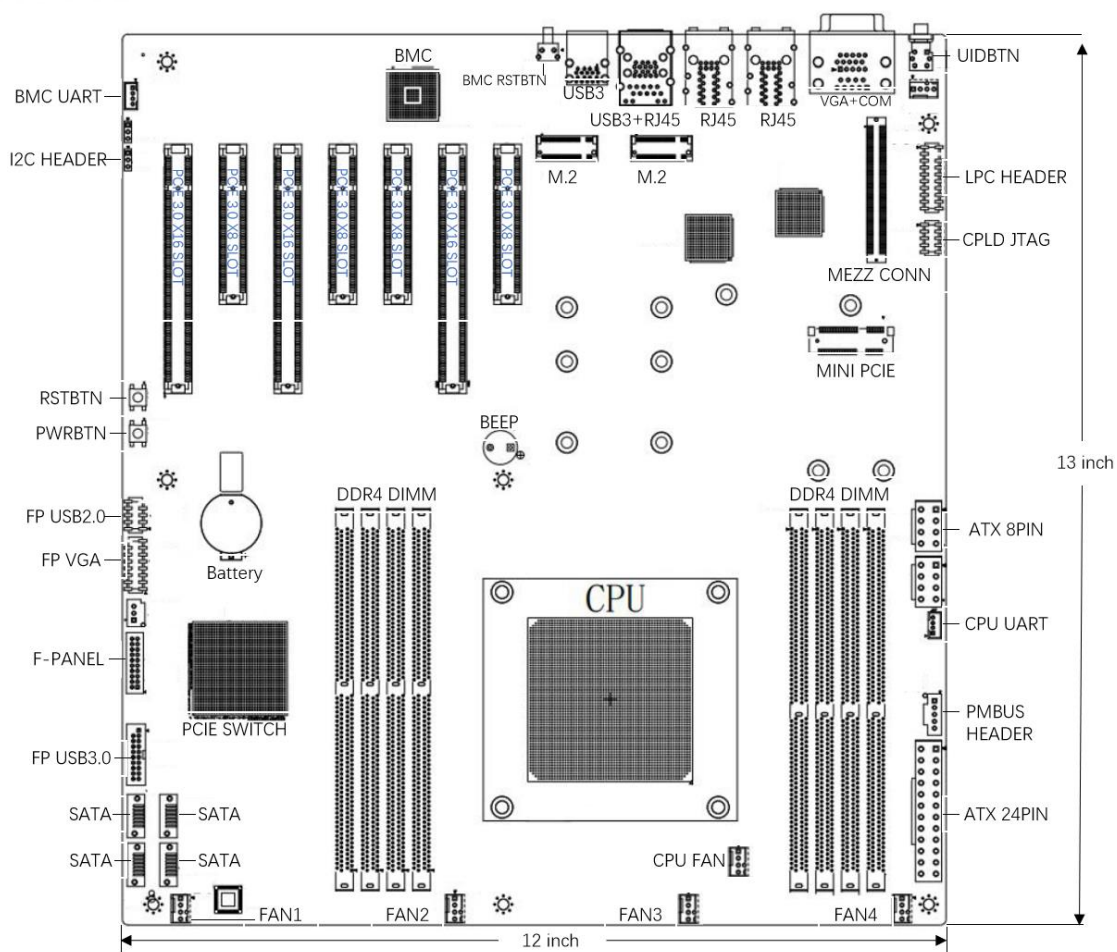


Figure 7.3-2 Mother Board Layout Diagram

7.3.6.2 Chassis

The 2U rack mount chassis contains the layout of the power supply which support 2 CRPS power supply, Hard disk drive and Solid State Drive (SSD) and fans. The chassis dimension is showed in following figure.

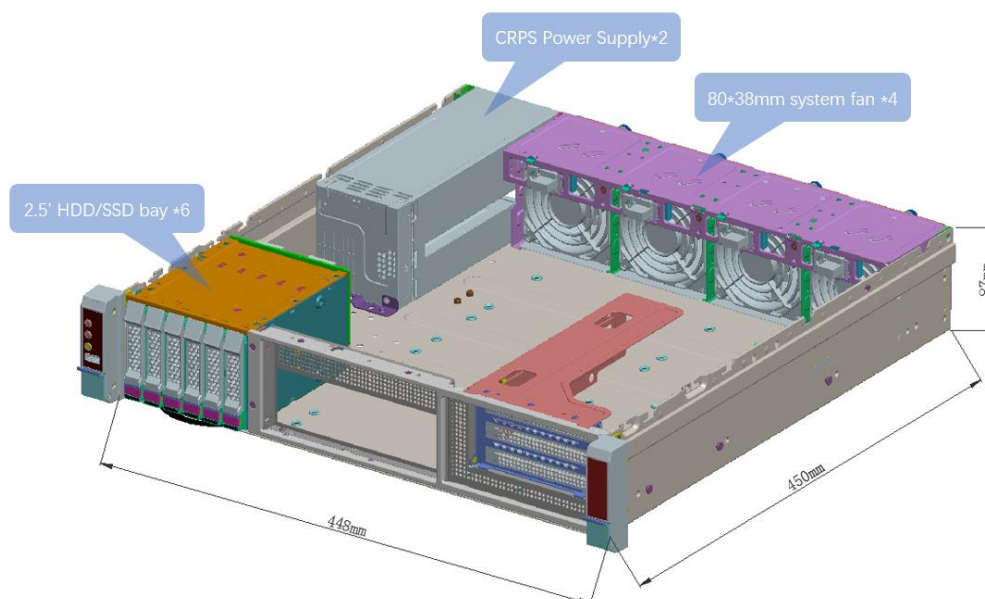


Figure 7.3-3 Chassis Mechanical Diagram

7.3.6.3 Cooling

The system installs 4* 80x38mm PWM fans for the cooling. (Support 4 pcs 80x38 fan is standard configuration, and 4 pcs 80*56mm system as optional)

7.3.7 Power Unit

In a fully loaded system which 3 PCIe slots, which contain 1 PCIe X16 slot support 75W power is standard power configuration, and system power supply have additional power 8pin/6pin cable to support special PCIe device which need more than 75W, and the other 2 PCIe X8 slot can support 25W is compliant to standard PCIe Criteria.

1. The X16 PCIe card support full high full long dimension. (DWFHFL)
2. The 2 X8 PCIe card support full high half long dimension (SWFHHL),
3. Additional, if X16 card is double size, there only one PCIe X8 supported.
4. The Unit support 2 CRPS 800W high-efficiency Power supply, redundancy design.

The O-DU_{7-2x} chassis includes two 550W/800W CRPS power supply unit (Support 1+1 redundancy design). The power supply unit is auto-switching capable, which enables it to automatically sense and operate at a 100v to 240v input voltage. The power supply unit features are listed in the following table.

Table 7.3-4 Power supply unit feature list

Item Name	Description
Output connectors	24pin/8pin/4pin/I2C
Dimension (W x D x H)	77 x 225 x 84 mm(CRPS criteria)
Maximum Output Power	+3.3V: 25A +5V: 25A +12V: 65A -12V: 0.5A +5Vsb: 3A
Rated Input Voltage/Current	100-240Vac / 10-5A
Rated Input Frequency	50/60HZ
Inrush current	Less than 30A

7.3.8 Thermal

Active cooling with up to 4 fans is integrated in the chassis.

The hardware acceleration cards use passive cooling and a custom heatsink and is equipped with temperature sensors. It is designed to operate in temperatures ranging from -5°C to +55°C.

7.3.9 Environmental and Regulations

The O-DU_{7-2x} hardware system is RoHS Compliant. The power supply unit is EMC FCC/CISPR Class B compliant. Table 7.3-5 lists the environmental related features and parameters.

Table 7.3-5 Environmental Features

Item Name	Description
Operating Temperature	-5°C to +45°C
Non-operating Temperature	-20°C to +60°C
Operating Relative Humidity	8% to 90% (non-condensing)

Non-operating Relative Humidity	5% to 95% (non-condensing)
---------------------------------	----------------------------

The hardware accelerator is designed to operate in indoor environments and in temperatures ranging from -5°C to +55°C.

7.4 O-RU_{7-2x} Hardware Reference Design

7.4.1 O-RU_{7-2x} High-Level Functional Block Diagram

Figure 7.4-1 provides a high-level functional block diagram depicting the major HW/SW components. It also highlights the internal/external interfaces that are required.

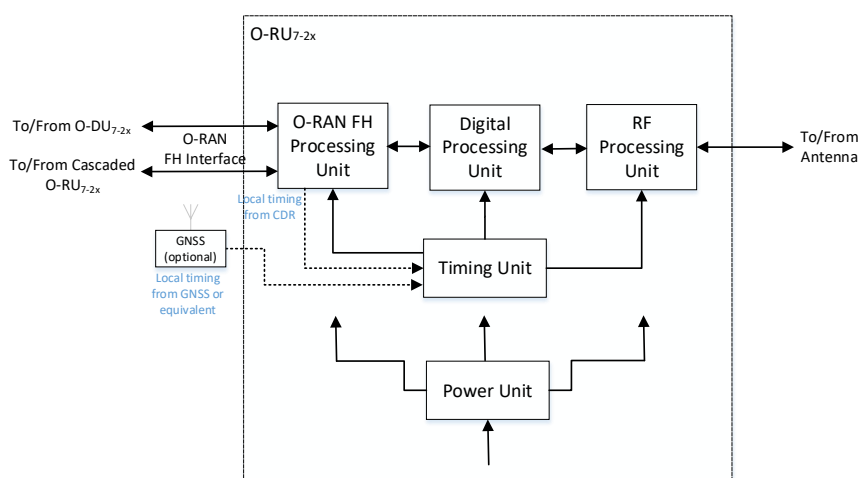


Figure 7.4-1 O-RU_{7-2x} High-Level Functional Block Diagram

7.4.2 O-RU_{7-2x} Hardware Components

7.4.2.1 O-RAN Fronthaul Processing Unit

The O-RAN Fronthaul Processing Unit receives eCPRI frames from the O-RAN fronthaul and process the control/data packets.

7.4.2.2 Digital Processing Unit

The digital processing unit of O-RU_{7-2x} is mainly for performing Low-PHY, DDC, DUC, CFR, DPD and synchronization. In this implementation, the ADC and DAC functions are also included in the digital processing unit.

This solution of the digital processing unit incorporates FPGA and a processor. The FPGA handles high speed digital processing, and the processor is used for hardware device configuration and the OAM function. The FPGA and the processor core can be integrated into one SoC or implemented into two devices. Here the FPGA and the processor core are integrated into one SoC device.

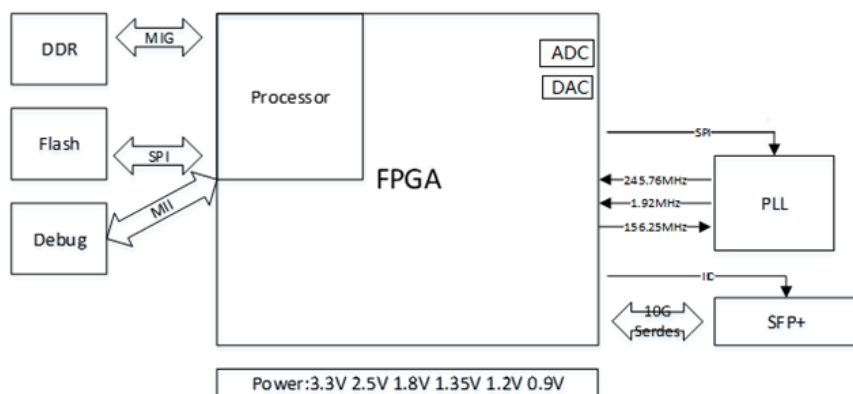


Figure 7.4-2 Digital Processing Unit Block Diagram

The FPGA resource requirements are shown in Table 7.4-1.

Table 7.4-1 FPGA Resource Requirements

System Logic cells	489 K
CLB Flip-Flops	447 K
CLB LUT	223 K
Distributed RAM	6.9 Mb
Block RAM	22.8 Mb
UltraRAM (Mb)	45 Mb
DSP Slices	1872
GTY Transceivers	8

The RF requirements of O-RU_{7-2x} are listed in Table 7.4-2

Table 7.4-2 RF Requirements

Parameter	Symbol	Min	Typ	Max	Unit	Test Condition/Comment
Transmitters						
Centre Frequency		650		6000	MHz	
Transmitter Synthesis Bandwidth				450	MHz	
Transmitter Large Signal Bandwidth				200	MHz	
Transmitter Attenuator Power Control Range		0		32	dB	Signal to noise ratio (SNR) maintained for attenuation between 0 and 20 dB
Transmitter Attenuation Power Control Resolution			0.05		dB	20 MHz LTE/NR at -12 dBFS

Parameter	Symbol	Min	Typ	Max	Unit	Test Condition/Comment
Adjacent Channel Leakage Ratio (ACLR)			-66		dB	75 MHz < f ≤ 2800 MHz
In Band Noise Floor			-154.5		dBm/Hz	0 dB attenuation; in band noise falls 1 dB for each dB of attenuation for attenuation between 0 dB and 20 dB
Out of Band Noise Floor			-153		dBm/Hz	600 MHz < f ≤ 3000 MHz
Maximum Output Power			6		dBm	0 dB attenuation; 3 × bandwidth/2 offset
Third Order Output Intermodulation Intercept Point	OIP3		27		dBm	600 MHz < f ≤ 3000 MHz
Error Vector Magnitude (3GPP Test Signals)	EVM					
1900 MHz LO			0.6		%	50 kHz RF PLL loop bandwidth
3800 MHz LO			0.53		%	300 kHz RF PLL loop bandwidth
Observation Receivers	ORx					
Center Frequency		450		6000	MHz	
Gain Range			30		dB	IIP3 improves dB for dB for the first 18 dB of gain attenuation; QEC performance optimized for 0 dB to 6 dB of attenuation only.
Analog Gain Step			0.5		dB	For attenuation steps from 0 dB to 6 dB
Receiver Bandwidth				450	MHz	
Maximum Usable Input Level	P _{HIGH}		-11		dBm	0 dB attenuation; increases dB for dB with attenuation.
Integrated Noise			-58.7		dBFS	450 MHz Integration bandwidth
			-57.5		dBFS	491.52 MHz Integration bandwidth
Third Order Input Intermodulation Intercept Point	IIP3					Maximum observation receiver gain; test condition: P _{HIGH} -11 dB/tone
Narrow Band						
1900 MHz			15		dBm	
2600 MHz			16.5		dBm	
3800 MHz			18		dBm	
Wide Band						IM3 products > 130 MHz at baseband; test condition: P _{HIGH} - 11 dB/tone;

Parameter	Symbol	Min	Typ	Max	Unit	Test Condition/Comment
1900 MHz			13		dBm	491.52 MSPS
2600 MHz			11		dBm	
3800 MHz			13		dBm	
Third Order Intermodulation Product	IM3		-70		dBc	600 MHz < f ≤ 3000 MHz
Spurious Free Dynamic Range	SFDR		64		dB	Non IMx related spurs, does not include HDx; (P _{HIGH} - 11) dB input signal
Harmonic Distortion						(P _{HIGH} - 11) dB input signal
Second Order Harmonic Distortion Product	HD2		-80		dBc	In band HD falls within ±100 MHz
			-73		dBc	Out of band HD falls within ±225 MHz
Third Order Harmonic Distortion Product	HD3		-70		dBc	In band HD falls within ±100 MHz
			-65		dBc	Out of band HD falls within ±225 MHz
Receivers						
Center Frequency				6000	MHz	
Analog Gain Step			30		dB	Attenuator steps from 0 dB to 6 dB
			1		dB	Attenuator steps from 6 dB to 30 dB
Receiver bandwidth				200	MHz	
Maximum Usable Input Level	P _{HIGH}		-11		dBm	0 dB attenuation; increase dB for dB with attenuation; continuous wave = 1800 MHz; corresponds to -1 dBFS at ADC 75 MHz <f≤3000 MHz
Noise Figure	NF		12		dB	0 dB attenuation at receiver port 600 MHz <f≤3000 MHz
Input Third Order Intercept Point	IIP3					
Difference Product	IIP3, d					Two (P _{HIGH} - 9) dB tones near band edge
2600 MHz (Wideband)			17		dBm	
2600 MHz (Midband)			21		dBm	
Sum Product	IIP3, s					Two (P _{HIGH} - 9) dB tones, at bandwidth/6 offset from the LO
2600 MHz (Wideband)			20		dBm	

Parameter	Symbol	Min	Typ	Max	Unit	Test Condition/Comment
HD3	HD3		-66		dBc	($P_{\text{HIGH}} - 6$) dB continuous wave tone at bandwidth/6 offset from the LO 600 MHz $< f \leq 4800$ MHz

7.4.2.3 RF Processing Unit

7.4.2.3.1 Power Amplifier (PA) Reference Design

The Power Amplifier boosts the RF output to the level required for the base station class.

7.4.2.3.1.1 Hardware Requirements

- Interface Requirements

The interface requirements of the transmit PA are listed in Table 7.4-3

Table 7.4-3 Power Amplifier Interface Requirements

Item Name	Description
RF Enable	The enable input should be compatible with 1.8V logic and tolerate 3.3V as required. A logic high enables the PA. A logic low disables the device and places it in a minimum dissipation mode.
RF Output	RF outputs support 50-ohm single ended to properly interface to a directional coupler, isolator, switch or antenna.
RF Input	RF inputs should support 50-ohm, single ended match to the transceiver output or preamp.

- Power Requirements

RF output power required is nominally 5 W (37 dBm). DC power is driven by efficiency of the PA and should not be more than 17W and the efficiency should be greater than 30%.

- RF Requirements

The PA and driver should have enough gain to boost the transceiver output to the rated power level. The output power should be at least 5W (37 dBm) including the loss of the circulator and antenna filter. The ACLR with DPD and CFR should be better than 47dBc according to the related 3GPP test models.

Table 7.4-4 Power Amplifier RF Requirements

Item Name	n 78
Gain	>60 dB including driver
P3dB	47 dBm
Input Return Loss	<-15 dB
Output Return Loss	<-15 dB
Switching Speed	<1uS
HD2	>28 dBc
HD3	>30 dBc

7.4.2.3.1.2 Hardware Design

RFin is the input of the PA, RFout is the output of the PA. Vdd and Vbias are the power input of the PA. RFen is the control pin to disable or enable the PA. The input and output should be matched to 50 ohms to optimize return loss as much as possible. Proper decoupling shall be provided to minimize impact of supply ripple on the RF output.

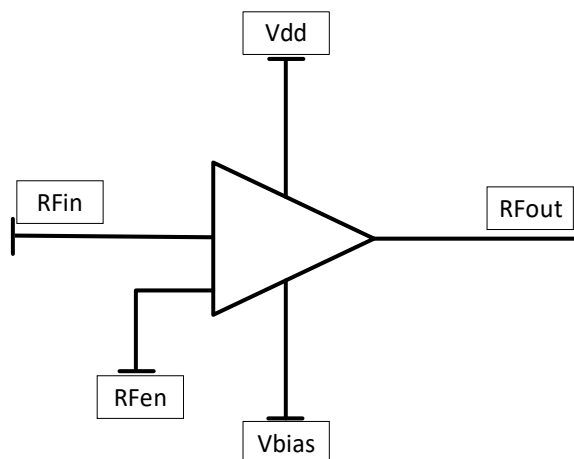


Figure 7.4-3 Power Amplifier Reference Design

7.4.2.3.2 Low Noise Amplifier (LNA) Reference Design

7.4.2.3.2.1 Hardware Requirements

- Interface Requirements

The interface requirements of the transmit LNA are listed in Table 7.4-5.

Table 7.4-5 LNA Interfaces Requirements

Item Name	Description
Enable	The enable input should be compatible with 1.8V logic and tolerate 3.3V as required. A logic high enables the LNA. A logic low disables the device and places it in a minimum dissipation mode.
LNA Bypass	The LNA Bypass skips the second stage LNA effectively reducing the overall gain. This would be selected when the input signal compromises the linearity of the amplifier.
RF Out	RF outputs support 50-ohm single ended to properly interface to the RF filtering and Rx input port of transceiver.
TR Switch	The TR switch is used to shunt any reflected RF power that reaches the LNA to a shunt load during the transmit period.
RF Input	RF inputs should support 50-ohm, single ended match to circulator.
Termination	This RF output port shunts any reflected transmit power to an appropriate load during the Transmit period to protect the input of the LNA.

- Power Requirements

DC power per channel should not exceed 0.5W.

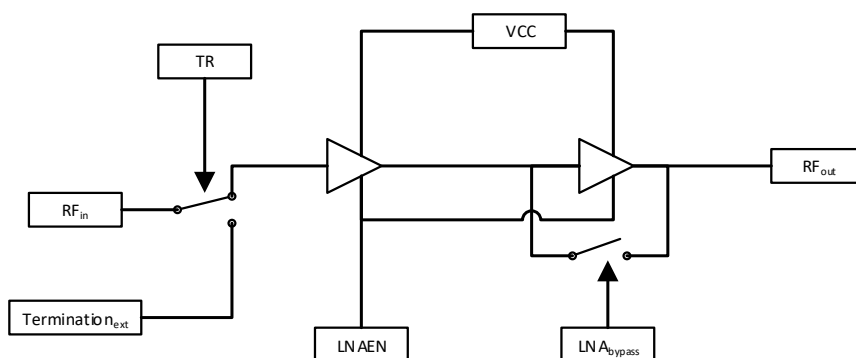
- RF Performance Requirements

Table 7.4-6 LNA RF Requirements

Item Name	n 78
Gain (High/Low)	34/17 dB
NF (High/Low)	1.6/1.6 dB
OIP3 (High/Low)	34/34 dBm
OP1dB (High/Low)	19/15 dBm
TR Switch Time	<1uS

7.4.2.3.2 Hardware Design

Figure 7.4-4 shows the configuration for a dual-stage LNA. The RF inputs and outputs should be properly matched to optimize the noise performance and flatness of the amplifier. The input switch is used to forward any reflected transmit energy to an external load during the transmit period to protect the LNA from damage. The DC supply voltage, VCC, should be properly filtered to prevent power supply noise from entering the LNA and reducing performance.


Figure 7.4-4 LNA Reference Design

7.4.2.3.3 Circulator Reference Design

For TDD use, the Tx and Rx links work in time duplex. To properly couple the Tx energy to the antenna and the Rx signal to the LNA input, a circulator is used to perform this function given the potentially high RF power being processed through this device. The circulator can be configured as a clockwise or counterclockwise device depending on the layout details.

7.4.2.3.3.1 Hardware Requirements

- Interface Requirements

The circulator includes 3 ports that can function as in input or an output. If the device is labelled clockwise, then for each input, its nominal output is the next pin rotating right around the device. If it is counterclockwise, the nominal output is the next pin rotating to the left around the device.

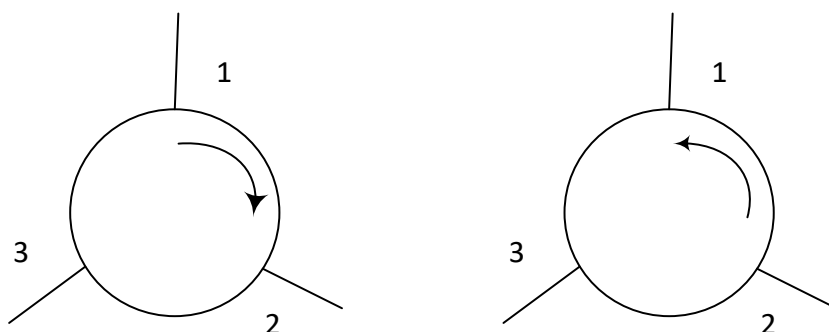
- RF Requirements

Table 7.4-7 Circulator RF Requirements

Item Name	n 78
Rotation	CW or CCW
Average Power	Up to 43 dBm
Peak Power	50 dBm
Impedance	50 ohms
Insertion Loss	-0.3 dB
Isolation	20 dB
Return Loss	22 dB

7.4.2.3.3.2 Hardware Design

The example on the left in Figure 7.4-5 is a clockwise oriented device. The RF input enters pin 1 and exits pin 2. Pin 2 can also be used as an input and its output is pin 3. Finally pin 3 can also function as an input with pin 1 being the output. The device on the right is counterclockwise oriented and the signal flow is the in the opposite direction.


Figure 7.4-5 Circulator Reference Design

7.4.3 Synchronization and Timing

The purpose of the clocking device is to accept the network reference clock, typically 1 pps, and generate a jitter free reference clock(s) for other devices in the system. The clock is typically part of an IEEE 1588 implementation either controlled by an external stack or implemented in the clock device itself which could be part of the baseband implementation.

7.4.3.1 Hardware Requirements

- Interface Requirements

The interface requirements of the transmit clocking are listed in Table 7.4-8.

Table 7.4-8 Interface requirements of the clocking

Item Name	Description
Reference Clock Input	The clock device should receive a network reference clock from the FPGA/ASIC. This typically could be a 1pps, 10 MHz or other standard reference as determined by the specific implementation. More than one input is allowed that may be selected between when the reliability of one reference is compromised. Standard differential clocking should be used to preserve signal integrity.
Control	Control of the transceiver is by way of 3 or 4 wire SPI or IIC functioning as a slave. Support for 1.8V control is required and tolerance of 3.3V is preferred. Typically, the clock devices will be part of an IEEE1588v2 protocol loop controlled by way of this control interface or other GPIO lines as required by the hardware implementation.
Clock Outputs	One or more clock outputs are required to drive the digital device and transceiver clock

	inputs. Each output should be independently programmable in frequency to suit the application. The outputs should nominally be differential to preserve clocking edge and to maximize tolerance of high common mode signals.
RF Inputs	RF inputs should support 50 ohm or 100 ohms, single ended or differential signalling to match the transceiver output.

- Power Dissipation

Total dissipation should be about less than 2W.

- RF Requirements

Clocking RF requirements are given in the following table.

Table 7.4-9 Clocking RF requirements

Absolution Phase Noise, Dual Loop Mode - LVDS OUTPUT	Typical	Additional Information
$f_{OUT} = 122.88 \text{ MHz}$		Example is using an external 122.88 MHz VCXO (Crystek CVHD-950); reference = 122.88 MHz; channel divider = 10 or 1; PLL2 loop bandwidth (LBW) = 450 kHz.
10 Hz Offset	-86 dBc/Hz	
100 Hz Offset	-106 dBc/Hz	
1 kHz Offset	-126 dBc/Hz	
10 kHz Offset	-135 dBc/Hz	
100 kHz Offset	-139 dBc/Hz	
800 kHz Offset	-147 dBc/Hz	
1 MHz Offset	-148 dBc/Hz	
10 MHz Offset	-157 dBc/Hz	
40 MHz Offset	-158 dBc/Hz	
Absolute Phase Noise, Single Loop Mode – LVDS OUTPUT	Typical	
LVDS OUTPUT		
$f_{OUT} = 122.88 \text{ MHz}$		Example is using an external 122.88 MHz reference (SMA100A generator); reference = 122.88 MHz; channel divider = 10; PLL2 LBW = 450 kHz.
10 Hz Offset	-111 dBc/Hz	
100 Hz Offset	-113 dBc/Hz	
1 kHz Offset	-123 dBc/Hz	
10 kHz Offset	-135 dBc/Hz	
100 kHz Offset	-140 dBc/Hz	

800 kHz Offset	-147 dBc/Hz	
1 MHz Offset	-148 dBc/Hz	
10 MHz Offset	-157 dBc/Hz	
40 MHz Offset	-157 dBc/Hz	
CLOCK OUTPUT ADDITIVE TIME JITTER (BUFFER MODE)	Typical	Example is using an external 122.88 MHz source driving VCXO inputs (distribution section only, does not include PLL and VCO)
$f_{OUT} = 122.88$ MHz	79 fS	Integrated BW = 200 kHz to 5 MHz
	101 fS	Integrated BW = 200 kHz to 10 MHz
	140 fS	Integrated BW = 12 kHz to 20 MHz
	187 fS	Integrated BW = 10 kHz to 40 MHz
	189 fS	Integrated BW = 1 kHz to 40 MHz
	176 fS	Integrated BW = 1 MHz to 40 MHz

7.4.3.2 Hardware Design

For the clocking function, it is usually performed by a synchronization IC which may include one or more PLLs. It also can supply numbers of output port to support different frequency clocking.

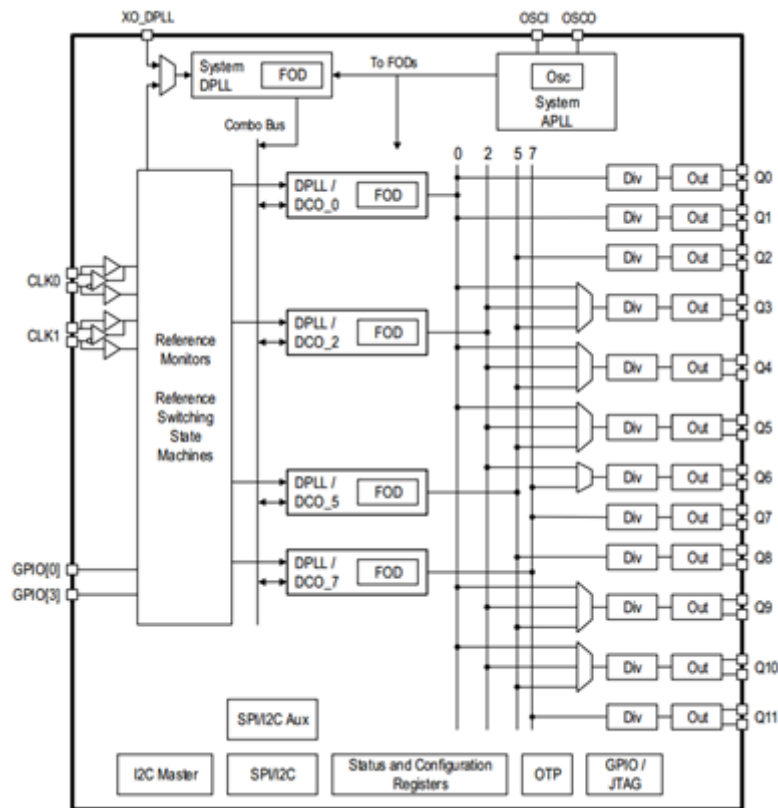


Figure 7.4-6 PLL Reference Design Diagram

CLK0 and CLK1 are differential reference clock inputs from the source to be cleaned up and used as the system reference. OSCI/OSCO should be connected to a crystal, and it also accepts a reference from a clock oscillator or a

fundamental mode parallel-resonant crystal. XO_DPLL is single-ended signal from the external TCXO. RESET(bar) is an active low pin to reset the internal logic to their default states. CS(bar) is an active low chip select for serial control. SCLK/SCL is a serial control port clock for both SPI and I2C. SDIO/SDA is a bidirectional serial data pin for both SPI and I2C. SDO is the serial data out pin for 4-wire mode. Q0 through Q11 are differential programmable output clock signals. These are the primary outputs of this device and provide high performance clock signals to the transceiver, baseband device and other key elements. This device is also responsible for providing SYSREF to various devices in the system if required. It may also use an external SYSREF as a source that may be retimed for local timing if necessary. GPIO0~9 are GPIO signals. The GPIO signals are intended to provide a user with a flexible method to manage the control and status of the part via pins without providing dedicated pins for each possible function that may be wasted in a lot of applications. The GPIOs are fully configurable so that any GPIO can perform any function on any target logic block.

7.4.4 External Interface Ports

The following table shows the external ports or slots provided by the O-RU_{7-2x}. External connections can be aggregated into a composite connector to minimize the points of environmental ingress as required for a specific deployment.

Table 7.4-10 External Port List

Port Name	Feature Description
Fronthaul interface	One 25Gbps optical interface to connect with O-DU _{7-2x}
	One 25Gbps optical interface for cascade
Debug interface	RJ45 for debug usage
Power interface	Two-pin male plug for power cable

7.4.5 Mechanical

One such mechanical structure is shown in Figure 7.4-7. The shell should provide appropriate environmental protection, RF shielding and passive thermal dissipation. Internally, thermal pads are used to contact local heat sources and transfer that heat to the frame which in turn dissipates the heat through bulk metal and fins used to increase the effective surface area. Other enclosures are possible.

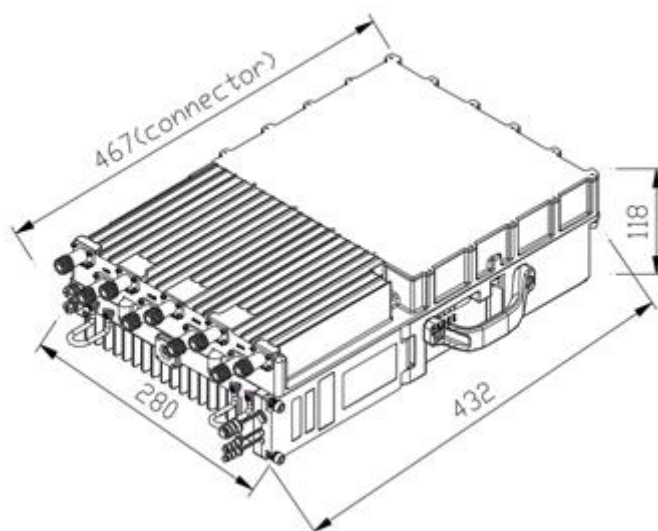


Figure 7.4-7 Mechanical Enclosure Diagram

7.4.6 Power Unit

For the power solution of the O-RU_{7-2x} can be over power cable. And for the board power solution, LDO and DC-DC can be used.

7.4.6.1 Hardware Requirements

The power unit is responsible for isolating and converting the telco supply to appropriate voltages for distribution to individual supply domains in the O-RU_{7-2x}. The typical dissipation is outlined below. The RF front end is broken out separately since different deployment scenarios will require different frequencies and power levels resulting in significantly different total dissipation. Total Power is the sum of the two sections below.

Table 7.4-11 Digital Processing Unit Power Requirements

Module	8T8R(W)	Notes
FPGA	40	
PLL	1.5	
Ethernet PHY	1	
DDR	1	
SFP+ Module	1.5	1 module installed
Others	3	
Total device consumption	48	
O-RU _{7-2x} digital processing unit power consumption	56.5	on board power efficiency 0.85

Table 7.4-12 RF Processing Unit Power Requirements

Module	8T8R(W)	Notes
PA	150	75% Tx,Max RF. 5W rms output@100MHz
PA Drive Amp	12	
LNA	5	28% Rx
Others	3	Bias generators and controller
Total device consumption	170	
O-RU _{7-2x} RF processing unit power consumption	200	on board power efficiency 0.85

7.4.6.2 Hardware Design

The block diagram of the power unit is shown in Figure 7.4-8, which depict the power distribution to various functional modules such as PA board and FPGA/RF SoC board(Digital) along with approximate power requirements.

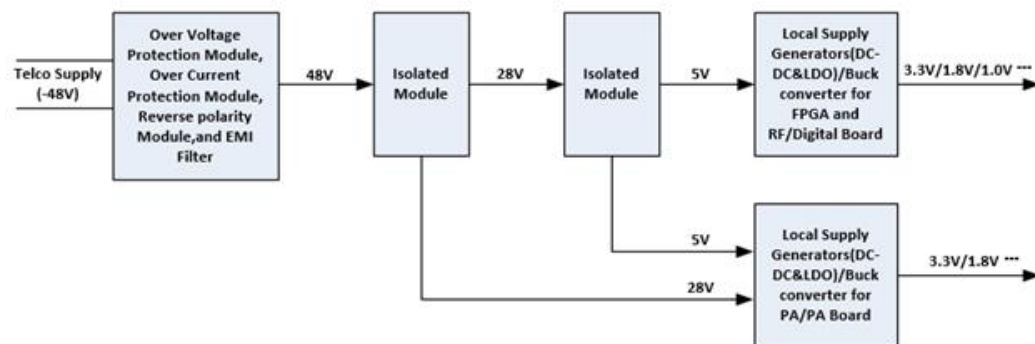


Figure 7.4-8 Power Unit Design

7.4.7 Thermal

The O-RU_{7-2x} will use passive cooling.

7.4.8 Environmental and Regulations

The O-RU_{7-2x} hardware system is RoHS Compliant.

Table 7.4-13 Environmental Features

Item Name	Description
Operating Temperature	-40°C to +55°C
Storage Temperature	-45°C to +70°C
Humidity	2% to 95%
Atmospheric Pressure	70kPa to 106kPa

History

Date	Revision	Description
2022.11.08	01.00.00	Published as final version 01.00